

11.1.2 Sum of Digits using Recursion

Algorithm

Step 1: Start

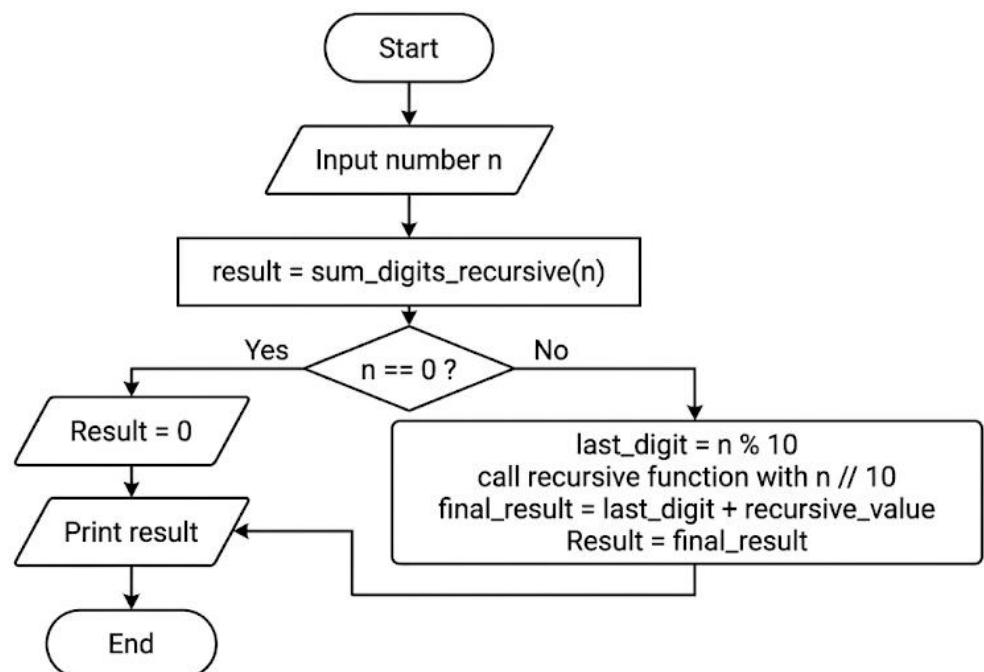
Step 2: Input number n

Step 3: Call function `sum_of_digits_recursive(n)`

Step 4: Check if $n == 0$

- If YES $\rightarrow$ return 0
 - If NO $\rightarrow$ go to next step
- Step 5: Calculate $(n \% 10)$ (last digit)
- Step 6: Call function again with $n // 10$
- Step 7: Add result $\rightarrow (n \% 10) + \text{recursive call}$
- Step 8: Return result
- Step 9: Print result
- Step 10: Stop

Flowchart



11.1.2. Sum of Digits using Recursion

Write a recursive function `sum_of_digits_recursive(n)` that calculates the sum of the digits of a given non-negative integer n .

Input Format:

- The input consists of a single non-negative integer n .

Output Format:

- The output should be the sum of the digits of the non-negative integer n .

Constraints:

- $0 \leq n \leq 10^9$

Sample Test Cases

+

SumofDi...

```

1  v
2  v
3  v
4  v
5  v
6  v
7  v
8  # Write your code here
9  number = int(input())
10 result = sum_of_digits_recursive(number)
11 print(result)

```

Debugger

Submit

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Average time

0.008 s

7.80 ms

Maximum time

0.011 s

11.00 ms

Test case 1

Expected output

23

Actual output

23

Test case 2

Expected output

5

Actual output

5

Terminal

Test cases

Prev

Reset

Submit

Next